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VoiceChess

An Audio-Based Chess Interface for Accessible Play

PROBLEM STATEMENT

Chess is a fundamentally visual and manual game. Traditional play depends on sighted recognition of the board and pieces, alongside fine motor control to physically move them. This excludes a significant population: individuals with visual impairments cannot independently track board state, and individuals with physical or motor disabilities may struggle to handle pieces or operate a keyboard and mouse for digital chess.

Existing accessible chess solutions, such as tactile boards with braille markings, are often expensive, require a sighted or trained assistant, and remain largely inaccessible to those with motor impairments as well. There is a clear need for a chess interface that removes both the visual and manual dependency entirely, relying solely on speech, so that independent play becomes possible regardless of physical ability. Voice Chess was conceived to directly address this gap.

ABOUT THE PROJECT

Voice Chess is an audio driven chess system that allows a player to conduct an entire game using only spoken language. A player states a move in natural chess notation, such as "knight to f3" or "pawn e4," and the system transcribes, parses, and validates it against the current board state. Every move made, whether by the player or their opponent, is announced aloud, allowing the full game to be followed without any visual reference.

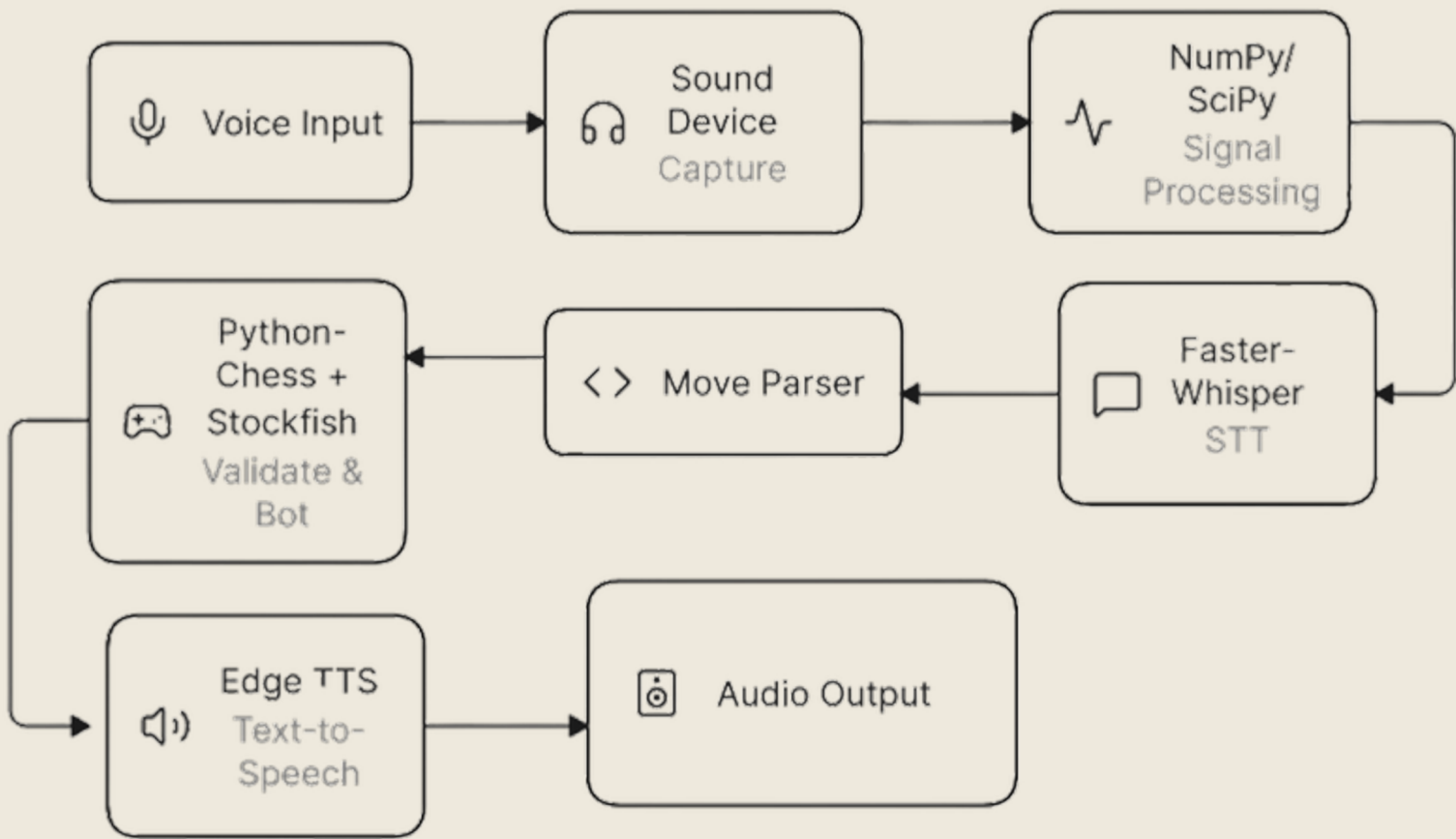
The system supports two distinct modes of play. The first is a manual dual player mode, where two individuals alternate speaking their moves and both receive audio confirmation of the resulting board state. The second is a single player mode against a computer opponent, which has now been fully implemented. This mode uses python-chess's UCI (Universal Chess Interface) integration to communicate with the Stockfish chess engine, which natively supports adjustable skill levels. This allowed multiple bot difficulty tiers to be incorporated into the voice controlled game loop with minimal additional engineering.

Architecturally, the system is lightweight and platform agnostic, requiring only a standard microphone and speaker. It is designed to run on laptops, mobile devices, or tablets without dependency on specialized hardware.

TOOLS USED

Tool / Library	Function
Faster-Whisper	Speech-to-text engine (an optimized reimplementation of OpenAI's Whisper) that transcribes the player's spoken move into text in real time.
Python-chess (Stockfish via UCI)	Core chess engine/library that maintains board state, validates legal moves, detects check/checkmate/stalemate, and interfaces with the Stockfish engine to power the single-player bot at multiple difficulty levels.
Microsoft Edge TTS (edge-tts)	Text-to-speech engine used to read moves, board state, and game events aloud. Primarily during testing/validation of the audio output pipeline.
SoundDevice	Captures live microphone audio input and handles audio playback output, bridging the physical microphone/speaker to the software.
asyncio	Manages asynchronous, non-blocking execution — allowing audio capture, transcription, and playback to run smoothly without freezing the game loop.
SciPy & NumPy	Handles numerical audio signal processing (e.g., array operations, resampling, filtering) on the raw audio data before it's passed to the speech recognizer.
keyboard	Provides manual input triggers (e.g., push-to-talk activation) to control when the system starts/stops listening.

PROCESSING PIPELINE



FEATURES

Independent operation: Removes the need for a sighted assistant or adapted physical equipment, enabling autonomous play.

Adjustable challenge: Multiple Stockfish backed difficulty levels allow a single player to practice against an opponent matched to their skill, rather than requiring a second human.

Immediate error correction: Illegal or misheard moves are flagged and communicated back to the player instantly, preventing silent errors in gameplay.

Hardware agnostic deployment: Functions across laptops, mobile phones, and tablets using only standard audio peripherals, keeping the barrier to entry low.

Cost efficient scalability: Built entirely on standard consumer hardware, avoiding the high cost typically associated with specialized accessible chess equipment.

FUTURE IMPROVEMENTS

Expand single-player mode with a wider range of bot difficulty levels, including an adaptive difficulty that adjusts to the player's skill over time.

Improve voice recognition accuracy for chess-specific vocabulary and accents, reducing misheard moves.

Add multilingual voice support so players can speak and receive moves in languages other than English.

Package the system into a lightweight, low-cost hardware device (e.g., Raspberry Pi + mic/speaker) for real-world deployment to visually impaired or physically disabled users.

Build and package a polished, accessible UI for general use — the current version is terminal-based, intended as an MVP for demonstration purposes only.